



Deepansh Reddy Ravula

Roll No.:240150030

B.Tech - Data Science and Artificial Intelligence

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	8.32 (Current)	2024-Present
Senior Secondary	CBSE Board	91.8%	2024
Secondary	CBSE Board	96.6%	2022

PROJECTS

- GradeOps : AI-Powered Exam Evaluation Platform** May. 2026 - Jun. 2026
Coding Club, IIT Guwahati [GitHub](#)
 - Developed a full-stack **academic assessment platform** using **React, FastAPI, and MongoDB**, enabling instructors to manage exams, students, marking schemes, and answer-sheet submissions through a unified dashboard.
 - Integrated a configurable **OCR pipeline** using **PaddleOCR** and **Qwen2-VL** for extracting text from scanned PDFs and images, supporting multi-page answer sheets, document preprocessing, and interchangeable OCR backends.
 - Built an automated grading workflow using **Google Gemini** to generate question-wise scores, evaluation feedback, and overall performance summaries, while implementing **JWT authentication** and persistent result tracking.
- CardioDetect : Deep Learning and Hybrid ML Pipeline** Mar. 2026 - May. 2026
AI Course Project, Professor Teena Sharma [GitHub](#)
 - Developed a **custom 3-block CNN** in PyTorch using Batch Normalization, LeakyReLU, and Dropout layers to classify ECG images into four cardiac conditions (MI, H.MI, AH, and Normal), achieving **97.31% accuracy**.
 - Implemented a **hybrid deep learning pipeline** by extracting feature embeddings from the CNN and training **SVM** and **Naive Bayes** classifiers, with CNN+SVM attaining a peak accuracy of **98.39% on ECG image classification**.
 - Leveraged the CNN as a **feature extractor** to generate 512-dimensional embeddings from ECG images, enabling integration of deep representations with machine learning classifiers for improved classification performance.
- Battery RUL Prediction : IPSO-PF Fusion Method** Jan. 2026 - Apr. 2026
Optimisation Course Project, Professor Amulya Kumar Mahto [GitHub](#)
 - Developed an **Improved Particle Swarm Optimization-Particle Filter (IPSO-PF)** framework with adaptive inertia weighting for robust lithium-ion battery capacity estimation, degradation tracking, and **RUL prediction**.
 - Achieved up to **89% reduction in capacity estimation MAE** compared to conventional Particle Filters on NASA battery datasets, reducing error from **0.0223 Ah to 0.0024 Ah** for accurate RUL estimation.
 - Validated the IPSO-PF approach on **NASA and CALCE battery datasets** using a dual-exponential degradation model, delivering accurate **Remaining Useful Life (RUL)** predictions and lower forecasting error than standard PF.

TECHNICAL SKILLS

- Programming Languages:** C/C++, Python, R*
 - Tech Stacks:** HTML, CSS, PHP, React.js*
 - Tools/Frameworks:** FastAPI, Scikit-learn, PyTorch*
 - Database Management:** MySQL, MongoDB
 - Operating Systems:** Windows, Linux*
 - Miscellaneous:** Git, Github
- * Elementary proficiency

KEY COURSES TAKEN

- Computer Science:** Data Structures and Algorithms(Theory and Lab), Database Management Systems(Theory and Lab), Python Programming(Lab), Introduction to Computing
- Data Science:** Introduction to Data Science, Introduction to Artificial Intelligence, Statistical Foundations for Data Science
- Mathematics:** Applied Probability and Random Processes, Introduction to Optimization, Discrete Mathematics

POSITIONS OF RESPONSIBILITY

- Core Team Member, Alcheringa, IIT Guwahati,** Apr. 2026 - Present
 - Worked as an Executive and Associate in the Public Relations and Branding team of Alcheringa
 - Supervised ideation, planning, and execution of ~50 competitions across India
 - Organized one of the largest cultural festivals in Northeast India, managing 10,000+ participants
- Core Team Member, Spirit, IIT Guwahati,** Apr. 2026 - Present
 - Managed social media content, promotional campaigns, and event coverage to enhance audience engagement and visibility for Spirit, IIT Guwahati.

ACHIEVEMENTS

- TS EAMCET**, Secured State Rank **412** among 3.5+ lakh candidates 2024

EXTRA CURRICULAR ACTIVITIES

- Volunteer, NSS, IIT Guwahati**, Engaged in various community service activities including cleanliness drives. 2026